Dra. Citlalli Almaguer Gómez

Professor–Researcher · Photonics, Visual Optics & Science Outreach

Department of Electro-Photonic Engineering · University Center for Exact Sciences and Engineering (CUCEI), University of Guadalajara · Zapopan, Jalisco · Mexico · citlalli.almaguer@academicos.udg.mx · ORCID: 0009-0003-3494-9580 · citlallialmaguerudg.github.io/personalweb
ResearchGate
National Researcher System · Candidate (SECIHTI) 2019–2023 · PRODEP Desirable Profile

PROFILE

I hold a PhD in Photonics and Laser Technologies from the University of Vigo (Spain) and a BSc in Physics from the University of Guadalajara. I am a professor–researcher in the Department of Electro-Photonic Engineering at the University Center for Exact Sciences and Engineering (CUCEI), University of Guadalajara. My research focuses on the design and validation of advanced optical elements —particularly coded phase plates and masks (cubic, trefoil and Jacobi–Fourier)— applied to the correction of complex refractive errors such as presbyopia, astigmatism, keratoconus and post-refractive-surgery conditions.

EDUCATION

2013 – 2017 **PhD in Photonics and Laser Technologies · Universidad de Vigo**

2011 – 2012 **MSc in Photonics and Laser Technology · Universidad de Vigo**

2004 – 2010 **BSc in Physics · Universidad de Guadalajara**

EXPERIENCE

2019 – present **Associate Research Professor B · Universidad de Guadalajara · CUCEI, Depto. de Ingeniería Electro-Fotónica**

PUBLICATIONS

2025 Astronomical Observation in Mexico Transitioning from the Visible to the Invisible: A Brief Historical Overview — *Óptica Pura y Aplicada 58(1)* · DOI

2025 Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations — *Atmosphere 16(5):628* · DOI

2025 Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios — *Climate 13(10):207* · DOI

2023 Characterization of a Shack–Hartmann Type Wavefront Sensor for Its Use in an Experimental Aberrometer — *Proceedings of SPIE* · DOI

2023 Caracterización de un divisor de haz para su utilización en un aberrómetro experimental — *SECIHTI / Centro de Investigaciones en Óptica*

2018 Pupil Size Stability of the Cubic Phase Mask Solution for Presbyopia — *Journal of Biomedical Optics 23(1):015002* · DOI

2017 Use of a Cubic Phase Plate as Solution for Moderate Astigmatism: Preliminary Study — *Óptica Pura y Aplicada 50(2)* · DOI

2017 Potential Use of Cubic Phase Masks for Extending the Range of Clear Vision in Presbyopes: Initial Calculation and Simulation Studies — *Ophthalmic and Physiological Optics 37(2)* · DOI

2017 Wavefront Coding for Visual Optics — *Proceedings of SPIE* · DOI

2017 Highly Aberrated Phase Elements for Presbyopia and Astigmatism Correction — *22nd Microoptics Conference (MOC'17), IEEE* · DOI

2016 Is Wavefront Coding an Alternative to Adaptive Optics for Retinal Imaging? — *15th Workshop on Information Optics (WIO), IEEE Xplore* · DOI

2015 Spreading Optics in the Primary School — *Journal of Physics: Conference Series 605:012040* · DOI

2014 Optics Activity for Hospitalized Children — *Proceedings of SPIE (ETOP)* · DOI

2014 The USC-OSA Student Chapter: Goals and Benefits for the Optics Community — *Proceedings of SPIE (ETOP)* · DOI

CONFERENCES & TALKS

2017 22nd Microoptics Conference (MOC'17) · Japan
2017 Third International Conference on Applications of Optics and Photonics (AOP) · Portugal
2016 15th Workshop on Information Optics (WIO) · Spain
2014 23rd Congress of the International Commission for Optics (ICO-23) · Spain
2014 7th European Meeting on Visual and Physiological Optics (EMVPO) · Poland

COMMUNITY BUILDING & SERVICE

2025 **Thesis supervision:** Demostración del efecto Rayleigh y Mie con pinzas ópticas (Advisor)
2025 **Thesis supervision:** Diseño de un microscopio funcional con fuente de iluminación LED para el análisis y procesamiento de imágenes 3D (Reviewer)
2025 **Thesis supervision:** Caracterización del desempeño de un prototipo de sensor de onda tipo Shack–Hartmann e incremento en la versatilidad de su software (Reviewer)
2025 **Thesis supervision:** Caracterización del desempeño de un prototipo de sensor de onda tipo Shack–Hartmann e incremento en la versatilidad de su software (Reviewer)

Review & assessment: Comité de Titulación · Ing. Fotónica, CUCEI–UDG, Comité Técnico · Ing. Informática, CUCEI–UDG
Project evaluation: External evaluator — 2020 Postdoctoral Stays in Mexico Call (2 applications)

RESEARCH INTERESTS

Visual optics & correction of refractive errors · Coded phase masks & plates (cubic, trefoil, Jacobi–Fourier) · Wavefront coding · Aberrometry & Shack–Hartmann sensors · Photonic engineering & applied optics · Atmospheric sciences & climate change · Stratospheric dynamics & sudden stratospheric warmings · Atmospheric optics · Science outreach & education

LANGUAGES

Spanish — Native · English — Intermediate · Galician — Basic

TOOLBOX

MatLab · Python · SLM · Shack–Hartmann · CCD · LaTeX · HTML5 / CSS3 / JS · Google Apps Script · PhET